Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 11 [2011 P1]

Test Date: 11th May 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCE-2011-P1-Model

Time: 3:00 **IMPORTANT INSTRUCTIONS** **Max Marks: 240**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 7)	Questions with Single Correct Choice	3	-1	7	21
Sec – II(Q.N : 8 – 11)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III(Q.N : 12 – 16)	Questions with Comprehension Type (2 Comprehensions – 2 + 3 = 5Q)	3	-1	5	15
Sec – IV(Q.N : 17 – 23)	Questions with Integer Answer Type	4	0	7	28
Total				**23**	**80**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 24 – 30)	Questions with Single Correct Choice	3	-1	7	21
Sec – II(Q.N : 31 – 34)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III(Q.N : 35 – 39)	Questions with Comprehension Type (2 Comprehensions – 2 + 3 = 5Q)	3	-1	5	15
Sec – IV(Q.N : 40 – 46)	Questions with Integer Answer Type	4	0	7	28
Total				**23**	**80**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 47 – 53)	Questions with Single Correct Choice	3	-1	7	21
Sec – II(Q.N : 54 – 57)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III(Q.N : 58 – 62)	Questions with Comprehension Type (2 Comprehensions – 2 + 3 = 5Q)	3	-1	5	15
Sec – IV(Q.N : 63 – 69)	Questions with Integer Answer Type	4	0	7	28
Total				**23**	**80**

CHEMISTRY: **Max.Marks: 80**

SECTION – I

(SINGLE CORRECT CHOICE TYPE)

This section contains **7 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

1. Among the following bidentate ligands, which one form more membered ring with metal ion in mononuclear complex?

A) $acac^{-1}$ **B)** $C_2O_4^{2-}$ **C)** $o-phen$ **D)** $dipy$

2. The correct order of increasing oxidizing power in the following is:

A) $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$ **B)** $Cr_2O_7^{2-} < VO_2^+ < MnO_4^-$

C) $MnO_4^- < Cr_2O_7^{2-} < VO_2^+$ **D)** $Cr_2O_7^{2-} < MnO_4^- < VO_2^+$

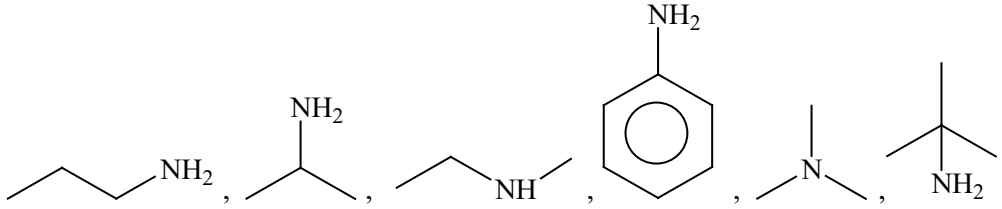
3.

S.NO	Polymer		Monomers		Uses
I)	Dacron	1)	Phenol, formaldehyde	P)	Making of combs, electrical switches
II)	Nylon 6,6	2)	Ethylene, glycol, pthalic acid	Q)	Paints and lacquers
III)	Glyptal	3)	Hexamethylene diamine Adipic acid	R)	Glass reinforcing material in safety helmets
IV)	Bakelite	4)	Ethylene glycol, terepthalic acid	S)	Sheets, bristles for brushes

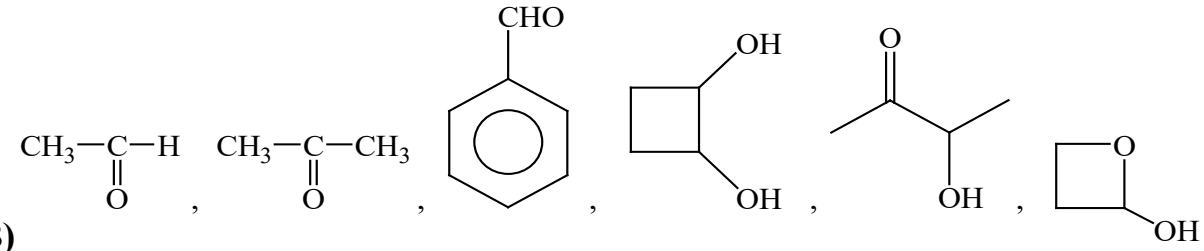
Then the incorrect combination from the above polymers is

A) I, 4, R **B)** III, 2, Q **C)** II, 3, S **D)** IV, 4, P

4. Which of the following reaction/statement is correct?

A) 

among these only four compounds will not give carbylamine test.

B) 

among these only four compounds gives tollen's test.

C) All $C_4H_{10}O$ isomers gives Lucas test.

D) All C_4H_8 isomers gives Baeyer's test.

5. A solution is prepared by dissolving 0.1 mole of Na_3PO_4 in sufficient water to make one litre solution. Calculate the concentration of OH^- ion in the solution.

Given: for H_3PO_4

$k_1 = 7.1\times10^{-3}, k_2 = 6.3\times10^{-8}, k_3 = 4.5\times10^{-13}$

A) $3.73\times10^{-2}\,M$ **B)** $4.70\times10^{-2}\,M$ **C)** $3.73\times10^{-3}\,M$ **D)** $4.70\times10^{-3}\,M$

6. Which of the following has ΔS_{sys} negative

A) Formation of non ideal solution showing negative deviation from the Raoult's law

B) Spontaneous rusting of iron

C) Conversion of diamond to graphite

D) Defect formation in a perfect crystal

7. The correct set of statements among the following

A) Micelles formation takes place above CMC and above kraft temperature

B) Larger the length of carbon chain of surfactant higher is the CMC at a given temperature

C) Addition of excess of electrolyte to sol suppresses the diffused double layer and reduces zeta potential there by causes coagulation

D) Tail of comet is visible due to Tyndall effect

A) A, B, C **B)** B, C, D **C)** A, B, D **D)** A, C, D

SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct

---

8. Which of the following combination(s) does not produce a paramagnetic gas?

A) XeF_4 and H_2O (complete hydrolysis)

B) Zn and dilute HNO_3

C) H_2O_2 and I_2 in basic medium

D) XeF_6 and H_2O (complete hydrolysis)

9. Which of the following order(s) is/are correct regarding bond dissociation enthalpies?(consider single bonds)

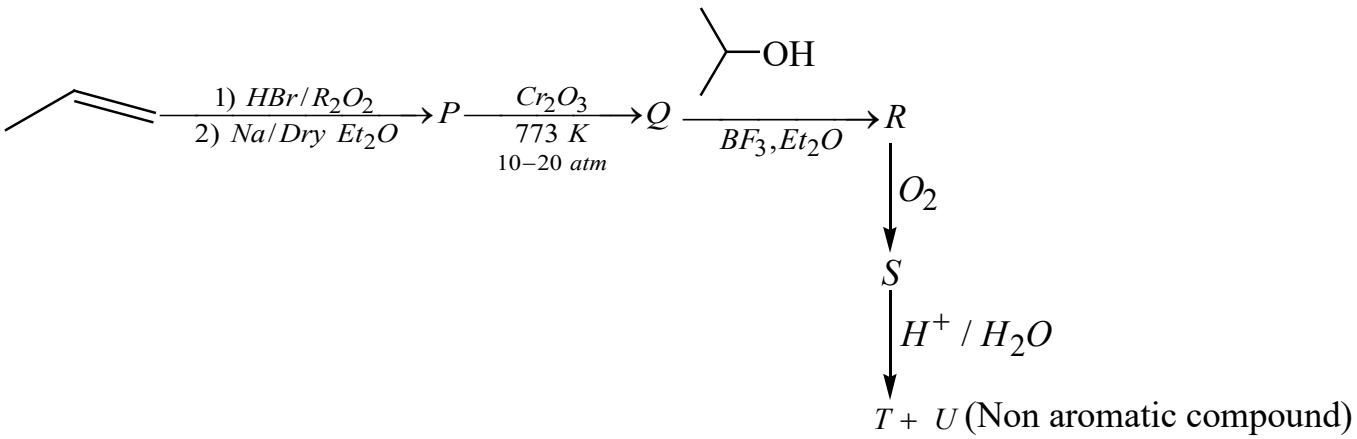
A) $C-C > Si-Si > Ge-Ge > Sn-Sn$

B) $P-P > N-N$

C) $S-S < O-O$

D) $Cl-Cl > Br-Br > F-F > I-I$

10.



Then the correct statements is/are

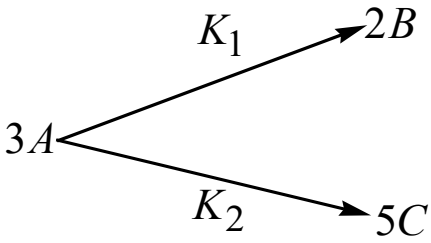
A) "U" gives 2,4 – DNP test.

B) Compound "T" reacts with phthalic anhydride in the presence of conc. H_2SO_4 gives a compound whose degree of unsaturation is "14".

C) Conversion of "Q" to "R" involves electrophilic aromatic substitution reaction.

D) "Q" decolourises pink colour of cold alkaline $KMnO_4$.

11. Select the correct statement(s) regarding the parallel reactions, in which both the reactions are of first order.



Given at t = 0, number of moles of A is 10; also given that $K_1 / K_2 = 3/2$.

Find the number of moles of B & C formed at $t = t_{1/2}(A)$

A) number of moles of B is 2.0 **B)** number of moles of B is 2.5

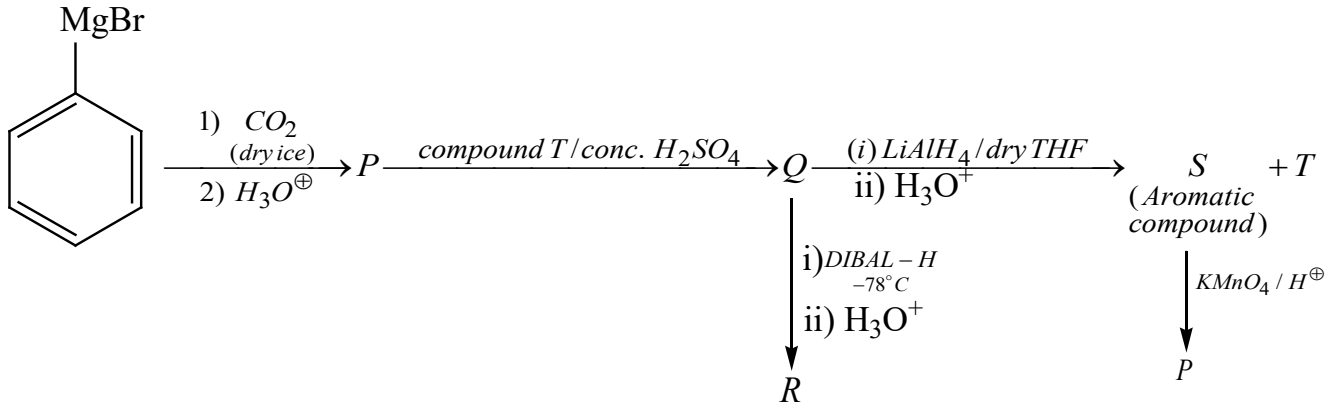
C) number of moles of C is 5/3 **D)** number of moles of C is 10/3

SECTION – III

(PARAGRAPH QUESTIONS TYPE)

This section contains 2 paragraphs. Based upon one of the paragraphs 3 multiple choice questions and based on the other paragraph 2 multiple choice questions have to be answered. Each of these questions has four choice (A) , (B) , (C) and (D) out of which ON:Y ONE is correct

Passage - I



12. Which of the following compound gives Lucus test?

A) Compound "P" **B)** Compound "Q"

C) Compound "S" **D)** Compound "R"

13. Which of the following compound gives 2, 4 – D.N.P test?

A) Compound "R" **B)** Compound "T"

C) Compound "P" **D)** Compound "Q"

14. Compound "S" and "T" can be differentiated by

A) Neutral $FeCl_3$ test **B)** Iodoform test

C) 2,4-DNP test **D)** Carbylamine test

Passage – II

When 306 g of $NH_4HS_{(solid)}$ is introduced in a 4.1 litre evacuated flask and heated to 227^0C then solid NH_4HS decomposes into gaseous ammonia and hydrogen sulphide. At equilibrium, it is found that 170 gm of $H_2S(g)$ are present in the container

15. If after establishment of equilibrium, H_2S is added in the container is such a way that partial pressure of H_2S now equals to original total pressure, then partial pressure of NH_3 at new equilibrium is

A) 51 atm **B)** 25 atm **C)** 100 atm **D)** 75 atm

16. If at 227^0C, ammonia also starts decomposing into nitrogen and hydrogen and establishes equilibrium $(2NH_3 \rightleftharpoons N_2 + 3H_2)$ then K_p for the reaction $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$ and total pressure developed in the container will-

A) decreases, increases **B)** increases, decreases

C) no change, no change **D)** no change, increases

SECTION –IV

(INTEGER ANSWER TYPE)

This section contains **7 questions.** The answer to each of the questions is a **single-digit integer,** ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS

17. How many complexes among the following are paramagnetic:

$\left[Mn(CN)_6\right]^{3-}$	$\left[Cr(H_2O)_6\right]^{3+}$	$\left[Co(en)_3\right]^{3+}$
$\left[V(CO)_6\right]$	$\left[Ni(NH_3)_6\right]^{2+}$	$\left[Ni(dmg)_2\right]$
$\left[PtCl_2(NH_3)_2\right]$	$\left[Cu(NH_3)_4\right]^{2+}$	$\left[Cu(CN)_4\right]^{3-}$

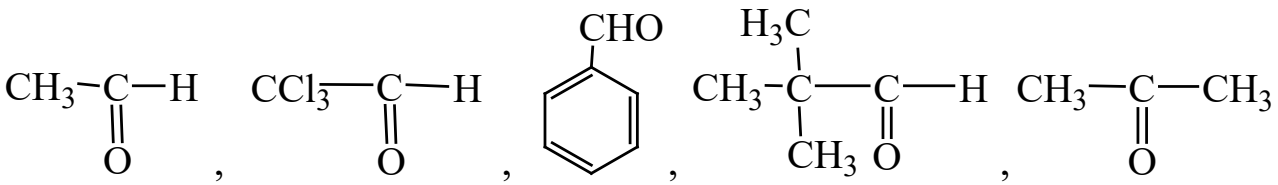
18. Number of monovalent oxygens present in the mineral $Ca_2Cu_2Si_3O_{10}.2H_2O$ is__

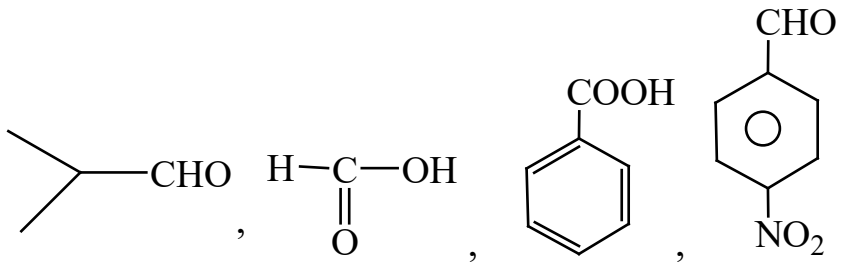
19. Total number of molecules having 'three centred two electorn' (3c-2e) bonds among the following:

B_2H_6, Al_2Cl_6, $BeCl_2(solid)$, $BeH_2(solid)$, $(AlH_3)_n$, $\left[Be(CH_3)_2\right]_n$, C_2H_6, $Al_2(CH_3)_6$, $C_2(CH_3)_6$

20. 2 : 1 mole ratio of benzaldehyde mixed with acetone in dilute NaOH and the mixture is continuously stirred for 30 minutes a pale yellow coloured solid compound is formed. Then the number of sp^3 hybridised carbon atoms in a pale yellow coloured compound is

21. How many of the following compounds can undergo Cannizaro reaction





22. 10 mole of liquid 'A' and 20 mole of liquid 'B' is mixed in a cylindrical vessel containing a piston arrangement. Initially a pressure of 2 atm is maintained on the solution. Now, the piston is raised slowly and isothermally. Assume ideal behaviour of solution, A and B are completely miscible.

$P_A^\circ = 0.6$ atm and $P_B^\circ = 0.9$ atm

Find out the pressure at which $\frac{1}{3}$rd of the total amount (by mol) of liquid solution taken initially, will be present in the vapour form. if the pressure is $\frac{119x}{300}$. find the value of x.

[GIVEN: $\sqrt{57} = 7.55$]

23. A metal crystallizes in such a lattice that only 60% of the total crystal space is occupied by the atoms. If atomic mass of the metal is 64π gram/mole and atomic radius is 0.2 nm. Find out density of metal in gram/cm^3?$\left[N_A = 6 \times 10^{23} \right]$

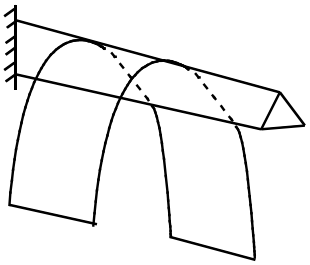
PHYSICS: **Max.Marks: 80**

SECTION – I

(SINGLE CORRECT CHOICE TYPE)

This section contains **7 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

24. A half thin pipe of mass 1kg and radius 20 cm is balanced on a knife edge. It's time period of small oscillations is: (assume no slipping at contact and $g = 10 m/\sec^2$)



A) $\frac{2\pi}{5}\sec$ **B)** $\frac{\pi}{5}\sec$ **C)** $2\pi\sqrt{\frac{\left(1-\frac{2}{\pi}\right)}{10}}\sec$ **D)** $\frac{\pi}{3}\sec$

25. For progressive wave: $y = (2cm)\sin(3t - 4x)$

Statement-1: The particle at $x = 0$ is moving towards positive y-axis at $t = 0$.

Statement-2: The particle at $x = 8m$ is in phase with particle at $x = 1m$.

Statement-3: The acceleration of particle at $x = \frac{\pi}{8}m$ at $t = \frac{\pi}{6}\sec$ is increasing in magnitude.

Statement-4: The velocity of particle at $x = \frac{\pi}{6}m$ is in direction of propagation of wave at all times.

The CORRECT statement(s) are:

A) 1 **B)** 1 and 3 **C)** 1,2 and 3 **D)** 1,2,3 and 4

26. For a liquid moving in a cylindrical tube of radius R and length l, the pressure difference at the ends of the tube is related by poiseuille's law given by $\Delta P = \left(\frac{8\eta l}{\pi r^4}\right)Q$. For a tube having radius $r = 2.00cm \pm 0.01cm$, $l = 10.0cm \pm 0.1cm$, the error in measurement of pressure difference is:

(Take: $\eta = \pi\ N - s/m^2$, $Q = 1m^3/\sec$)

A) 2% **B)** 3% **C)** 4% **D)** 6%

27. A photoelectric material having work-function ϕ_0 is illuminated with light of wavelength $\lambda\left(\lambda < \frac{hc}{\phi_0}\right)$. The fastest photoelectron has a de-Broglie wavelength λ_d.

A change in wavelength of the incident light by $\Delta\lambda$ results in a change $\Delta\lambda_d$ in λ_d.

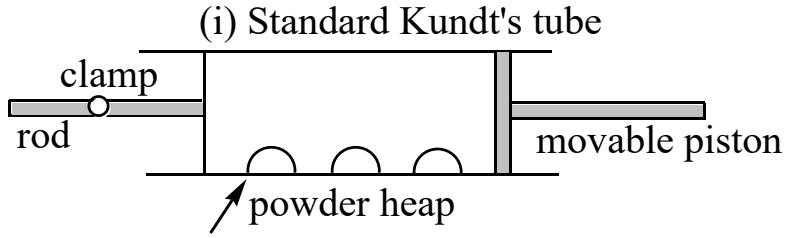
Then the ratio $\Delta\lambda_d / \Delta\lambda$ is proportional to

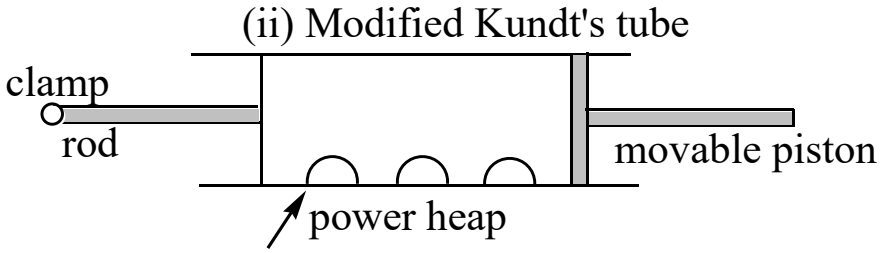
A) λ_d^3 / λ^2 **B)** λ_d^3 / λ **C)** λ_d^2 / λ^2 **D)** λ_d / λ

28. A plane electromagnetic wave of angular frequency ω propagates in a poorly conducting medium of conductivity σ and relative permittivity ε. Find the ratio of conduction current density and displacement current density in the medium.

A) $\frac{\varepsilon\varepsilon_0\omega}{\sigma}$ **B)** $\frac{\sigma}{\varepsilon\varepsilon_0\omega}$ **C)** $\frac{\omega}{\sigma\varepsilon_0\varepsilon}$ **D)** $\frac{\omega\sigma}{\varepsilon_0\varepsilon}$

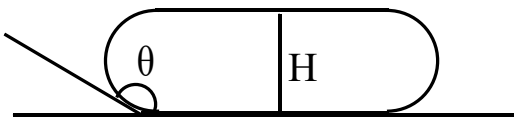
29. In the Kundts tube experiment (shown in fig.(i)), the rod is clamped at the end instead of clamping it at the center as shown in fig. (ii). It is known that speed of sound in air is 330 m/s, powder piles up at successive distance of 0.6 m and length of rod used is 1m, speed of sound in rod is (rod is in first hormonic)





A) 550 m/sec **B)** 1100 m/sec **C)** 1200 m/sec **D)** 600 m/sec

30. Consider a large drop of liquid on a glass plate. Contact angle is θ and density of liquid is ρ. Surface tension is S



A) $H = \sqrt{\frac{2S}{\rho g}(1-\sin\theta)}$ **B)** $H = \sqrt{\frac{2S}{\rho g}(1+\sin\theta)}$

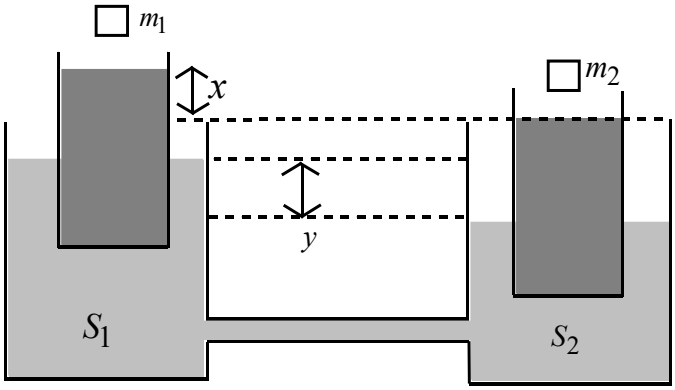
C) $H = \sqrt{\frac{2S}{\rho g}(1+\cos\theta)}$ **D)** $H = \sqrt{\frac{2S}{\rho g}(1-\cos\theta)}$

SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

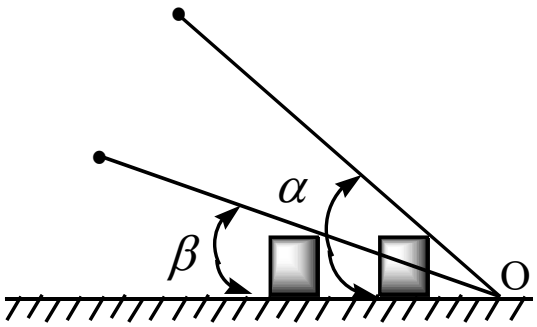
This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct

31. Two identical cylinders are connected by a pipe. In both cylinders we fill water partially and float two smaller cylinders as shown. The left cylinder has area S_1 & right one has area S_2. These cylinders are both massless. They are filled with equal volume of water (V). Now in left cylinder we float a block of mass m_1 and in right cylinder we float a block of mass m_2. Before floating, the level in floating cylinders differed by x_1 & level in lower cylinders differend by y_1. The same values after floating are x_2 & y_2 respectively.



A) $x_1 = V\left(\frac{1}{S_1} - \frac{1}{S_2}\right)$ **B)** $y_1 = 0$ **C)** $x_2 = \frac{(m_1 - m_2)(S_1 - S_2)}{\rho S_1 S_2}$ **D)** $y_2 = \frac{m_1 - m_2}{\rho S_2}$

32. A cubical block of mass m and side length b is placed on a smooth floor. A smooth and rigid rod of length L and with negligible mass is leaning against the block. A small sphere of mass M is attached to the upper end of the rod. The lower end of the rod is fixed at point-O, but the rod can rotate freely around the point within the vertical plane. Initially the angle between the rod and the floor is α while the system is at rest. After releasing, Choose correct statement(s)



A) Speed of the block when angle between rod and the floor is β is

$$b\sqrt{\frac{MgL(\sin\alpha - \sin\beta)}{mb^2 + ML^2\sin^4\beta}}$$

B) Speed of the block when angle between rod and the floor is β is

$$b\sqrt{\frac{2MgL(\sin\alpha-\sin\beta)}{mb^2+ML^2\sin^4\beta}}$$

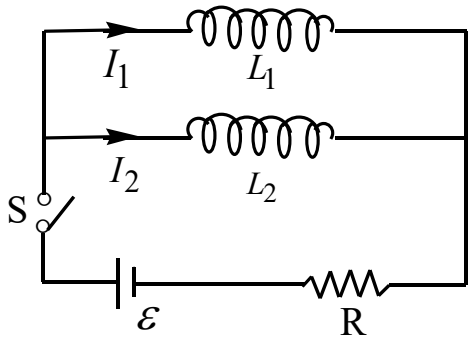
C) Angular speed of the rod when angle between rod and the floor is β is

$$\sqrt{\frac{2MgL(\sin\alpha-\sin\beta)}{mb^2\cos ec^4\beta+ML^2}}$$

D) Angular speed of the rod when angle between rod and the floor is β is

$$\sqrt{\frac{MgL(\sin\alpha-\sin\beta)}{mb^2+ML^2\cos^4\beta}}$$

33. In the circuit shown, the switch is closed at $t=0$ (L_1 and L_2 are ideal inductors and initial current at both inductors is zero)



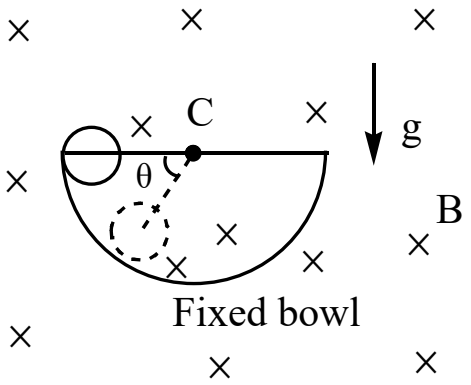
A) Just after $t=0, I_1=I_2=0$

B) at any time $t=t, \frac{I_1}{I_2}=\frac{L_2}{L_1}$

C) at any time $t=t, I_1+I_2=\frac{\varepsilon}{R}$

D) at $t=\infty$, current flowing through the resistor is $\frac{\varepsilon}{R}$.

34. A solid sphere is held at rest on the fixed hemispherical bowl made of insulating material such that centre of sphere lies on the horizontal diameter as shown in figure. Total charge $q=5.1\times10^{-1}C$ and mass $m=42$ grams, both are distributed uniformly in the sphere. There exists unform magnetic field B in space as shown in the figure. consider contact surface to be sufficiently rough. Spherical bowl is made of non-magnetic material. Radius of bowl = 10 cm, radius of sphere 3 cm. $g=10\,m/s^2$. Now if sphere is released



A) At an angle θ with horizontal, velocity of COM of sphere will be

$v = \sqrt{\frac{10}{7} g(R-r)\sin\theta}$.

B) At an angle θ with horizontal, velocity of COM of sphere will be

$v = \sqrt{\frac{8}{7} g(R-r)\sin\theta}$.

C) The limiting value of B (approximately) so that on releasing sphere, it do not leaves contact with bowl in course of motion will be 2T

D) The limiting value of B (approximately) so that on releasing sphere, it do not leaves contact with bowl in course of motion will be 4T

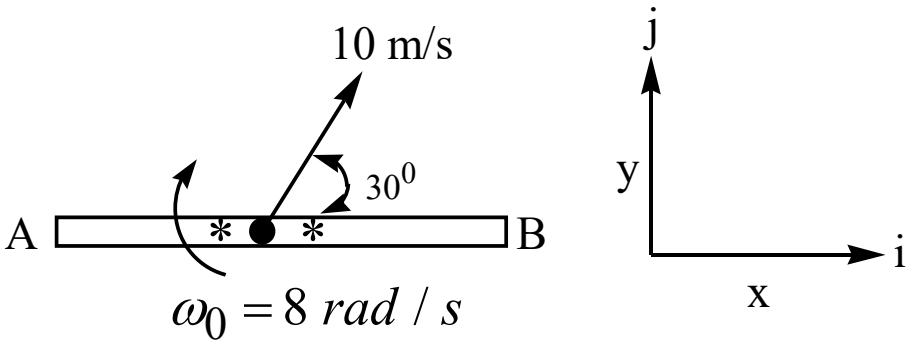
SECTION – III

(PARAGRAPH QUESTIONS TYPE)

This section contains 2 paragraphs. Based upon one of the paragraphs 3 multiple choice questions and based on the other paragraph 2 multiple choice questions have to be answered. Each of these questions has four choice (A) , (B) , (C) and (D) out of which ON:Y ONE is correct

Passage-1

A thin unform rod AB of mass M = 1 kg and length 1m is held horizontally with two small insects each of mass $m = \left(\frac{M}{6}\right)$ at centre of rod. At time $t = 0$ rod is projected in vertical x-y plane such that its initial velocity of projection of centre is 10 m/s at an angle 30^0 with horizontal and angular velocity of rod is $\vec{\omega}_0 = 8 rad/s\ (-k)$.



Immediately after projection both insects starts moving on rod (without slipping) away from centre is opposite directions with speed $\frac{1}{2} m/s$ with respect to rod. Take acceleration due to gravity $g = 10\ m/s^2$. Answer the following questions:

35. While rod is in motion in vertical plane under influence of gravity.

A) Sum of kinetic energy and gravitational potential energy of syste (insects + rod) will remain conserved.

B) Angular momentum of system (rod + insects) about centre of mass of system will remain conserved

C) linear momentum of system will remain conserved

D) Angular velocity of rod will remain constant.

36. Angle through which rod will rotate before centre of rod comes to same horizontal level as it was at the instant when rod was initailly projected, is

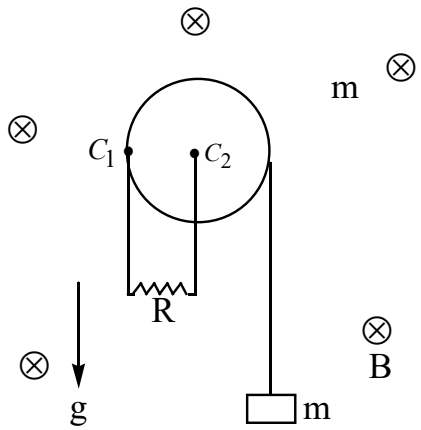
A) 3π radians **B)** 2π radians **C)** 8π radians **D)** none of these

37. Friction force acting on insects when they reach ends of rod without slipping

A) 6N **B)** $\frac{2}{3} N$ **C)** $\frac{4}{3} N$ **D)** $\frac{1}{6} N$

Passage-2

Conside a perfectly conducting unform disc of mass m and radius 'a' hinged in a vertical plane from its centre and free to rotate with respect to hinge. A resistance R is connected between centre of the disc and periphery by using two sliding contacts C_1 & C_2. A long non-conducting massless string is wrapped around the disk, whose another end is attached with a block of mass m. There exists a unform horizontal magnetic field B, whose arrangement is shown in figure.



Given system is released from rest at $t = 0$. Assume friction between string and disc is sufficient so that there is no slipping between them. Let at any instant t, velocity of block is v, angular velocity of the disc is ω and current in resistance is i.

38. Find the terminal speed of the block of mass m is (Terminal speed = constant speed attained by the block after very long time)

A) $\frac{mgR}{B^2a^2}$ **B)** $\frac{2mgR}{B^2a^2}$ **C)** $\frac{3mgR}{B^2a^2}$ **D)** $\frac{4mgR}{B^2a^2}$

39. Just after $t = 0$, the acceleration of block is:

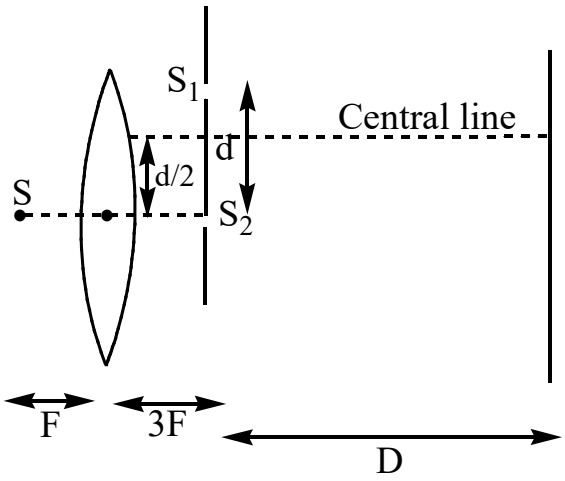
A) g **B)** $\frac{g}{2}$ **C)** $\frac{2g}{3}$ **D)** $\frac{g}{5}$

SECTION –IV

(INTEGER ANSWER TYPE)

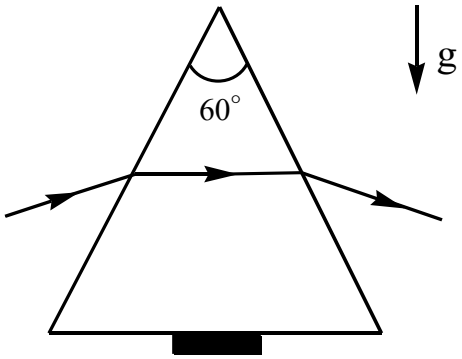
This section contains **7 questions.** The answer to each of the questions is a **single-digit integer,** ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS

40. A monochromatic source of light S is placed at focus of a coverging lens of focal length F. Two identical slits are at '3F' distance from the lens and its central line is $\frac{d}{2}$ distance above optical axis of lens. If D and d have their usual meaning, then the distance of the central maxima on the screen from central line $(D >> d >> \lambda)$ is $\frac{x\lambda D}{d}$ then fill the value of x.



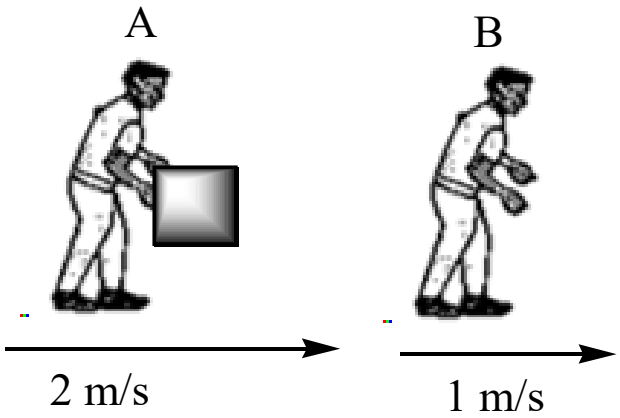
41. An equilateral glass prism kept on a table has refractive index of $\mu = \sqrt{2}$. It is illuminated by a narrow laser beam having power P_0 and wavelength λ. The path of the laser beam inside the prism is parallel to the base of the prism. The change in reading of weight of

the prism due to the incident laser beam is $\frac{P_0}{c}\left(\frac{\sqrt{x}-1}{\sqrt{y}}\right)$. Find $x+y$? c is the speed of light. (ignore any partial reflection)

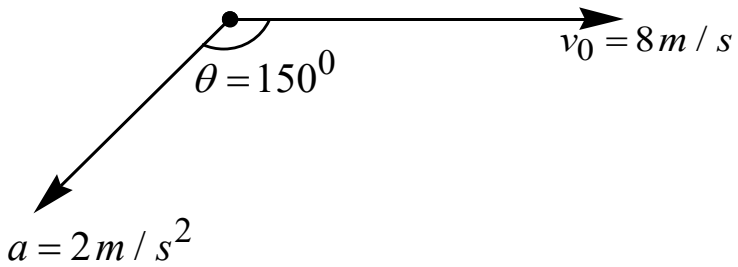


Weighing machine

42. Two astronauts, A and B, both with mass of 60 kg, are moving along a straight line in the same direction in a "weigthless" spaceship. Relative to the space ship the speed of A is 2 m/s and that of B is 1 m/s. A is carrying a bag of mas 5 kg with him. To avoid collision with B, A throws the bag with a speed v relative to the spaceship towards B and B catches it. The minimum value of v is $\left(x+\frac{76}{100}\right)m/s$ for which collision is avoided. Find the value of x. assume no change in velocity of spaceship.



43. The figure show the velocity and acceleration of point like body at the initial moment of its motion.The acceleration vector of the body remains constant.The minimum radius of curvature (in m) of trajectory of the body is



44. A rod of length L and mass M is having cross section A. The material of rod is having Young's modulus equal to 'Y'. If the rod is rotated horizontally about an axis passing through one end and perpendicular to length then ratio of elongation produced in the outer half of the rod to its inner half will be n:11, find n

45. A Hydrogen-like atom has atomic number Z. When it collides with stationary hydrogen atom in ground state, it needs a minimum KE of 51 eV to excite the hydrogen atom. Photons emitted in the electronic transitions from level n=4 to level n=2 in these atoms are used to perform photoelectric effect experiment on a target metal. The maximum kinetic energy of the photoelectrons generated is 4 eV . The photoelectric threshold wavelength for the target metal is closed to 100X nm where X is

46. The minimum kinetic energy needed by ${}_7N^{16}$ nucleus to cause the nuclear reaction ${}_7N^{16} + {}_2He^4 \rightarrow {}_1H^1 + {}_8O^{19}$ in a laboratory frame is n (in MeV). Assume that the alpha particle is at rest in laboratory frame. The masses of reaction ${}_7N^{16}, {}_2He^4, {}_1H^1$ and ${}_8O^{19}$ are 16.006u, 4.003u, 1.008u and 19.003u respectively where $u = 930\ MeV/c^2$. The value of n is close to _____

MATHEMATICS: **Max.Marks: 80**

SECTION – I

(SINGLE CORRECT CHOICE TYPE)

This section contains **7 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

47. The equation $\sin x = [1+\sin x] + [1-\cos x]$ has 'k' number of solutions in $[0,10\pi]$, then k is equal to (Here [.] represents greatest integer function)

A) 0 **B)** 10 **C)** 11 **D)** 20

48. The solution of $x^2\frac{dy}{dx} - xy = 1+\cos\frac{y}{x}$ is $\tan\left(\frac{y}{px}\right) = c - \frac{1}{qx^2}$ where c is constant, then the value of $p+q$ is

A) 3 **B)** 4 **C)** 5 **D)** 8

49. For $x = \frac{-17}{13}$, the value of $2\sin^{-1}\left(\frac{9-4x^2}{9+4x^2}\right) + 3\cos^{-1}\left(\frac{12x}{9+4x^2}\right) + \tan^{-1}\left(\frac{12x}{9-4x^2}\right)$ is

A) $\frac{5\pi}{4}$ **B)** 10π **C)** 5π **D)** $\frac{5\pi}{2}$

50. The value of the definite integral $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{dx}{(\sin x+\cos x+2\sqrt{\sin x\cos x})\sqrt{\sin x\cos x}}$ equals

A) 1 **B)** 2 **C)** $\frac{1}{2}$ **D)** $\frac{1}{4}$

51. Consider a function $f: A \to A$ on the set $A = \{1,2,3,4\}$, how many of such non-identical one-one functions satisfy the condition $|f(i)-i| < 2\ \forall i \in A$?

A) 4 **B)** 5 **C)** 6 **D)** 7

52. Consider the function $f: C \to C$ defined by $f(a+ib) = e^a(\cos b + i\sin b), a,b \in R$

Where $i = \sqrt{-1}$. Then which of the following is correct?

A) f is one – to – one and onto. **B)** f is one – to – one but not onto.

C) f is onto but not one – to – one **D)** f is neither one – to – one nor onto

53. The limit $\lim_{n\to\infty} n^{-\frac{3}{2}}\left((n+1)^{(n+1)}(n+2)^{(n+2)}.....(2n)^{(2n)}\right)^{\frac{1}{n^2}}$ equals

A) 0 **B)** 1 **C)** $e^{-\frac{1}{4}}$ **D)** $4e^{-\frac{3}{4}}$

SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct

54. If $2P(A \cup B) = 3P(A) = 4P(B) = 6P\left(\frac{A}{B}\right) \neq 0$, then which of the following is/are CORRECT? (where $P(x)$ denotes probability of occurrence of event x)

A) $P(A \cap \overline{B}) = \frac{1}{2}$ **B)** $P\left(\frac{B}{A}\right) = \frac{1}{4}$

C) A & B are independent events ` **D)** A & B are exhaustive events

55. Let $f : R \to (0,\infty)$ is a continuous function so that $f(x) + \frac{1}{f(x)} = e^x + e^{-x}\ \forall x \in R$, then which of the following is/are CORRECT?

A) number of such function(s) $f(x)$ is 1

B) number of such function(s) $f(x)$ is 2

C) area bounded by $y = f(x)$, x-axis, ordinate $x = -1$ and ordinate $x = 1$ can be $e - \frac{1}{e}$

D) area bounded by $y = f(x)$, x-axis, ordinate $x = -1$ and ordinate $x = 1$ can be $2e - 2$

56. If a_1, a_2, a_3, a_4, a_5 are distinct positive terms in A.P. having common difference d, then which of the following is/are CORRECT?

A) $5a_3^2 > 4d^2$

B) sum of all terms $= 5a_3$

C) $a_1 + 5a_5, 3a_3, 2a_2 + 4a_4$ are in A.P.

D) $a_1 a_5 < a_2 a_4$

57. Let 'P' be a point on $y^2 = 4x$ which is at shortest distance from centre of circle $x^2 + y^2 - 4x - 16y + 64 = 0$ and Q is point of intersection of circle and line segment joining point P and centre of circle then which of the following is/are CORRECT?

A) slope of tangent of circle at point Q is equal to $\left(\frac{1}{2}\right)$

B) distance between centre of circle and point P is $2\sqrt{5}$

C) co-ordinates of point P is (4, 4)

D) x-intercept made by normal at point P is 4

SECTION – III

(PARAGRAPH QUESTIONS TYPE)

This section contains 2 paragraphs. Based upon one of the paragraphs 3 multiple choice questions and based on the other paragraph 2 multiple choice questions have to be answered. Each of these questions has four choice (A) , (B) , (C) and (D) out of which ON:Y ONE is correct

Passage – 1

Two teachers A and B write questions and corresponding solutions for a mathematics test. A writes 10 questions every hour but makes the mistakes in 10% of the solutions. B writes 20 questions per hour and makes the mistakes in the 20% of the solutions. Both of them work for 3 hours and send all the questions to teacher C to be checked. However, teacher C is not really so smart and only 75% of the questions he thinks are wrong are actually incorrect.

Teacher C thinks 20% of questions from A have incorrect solutions and that 10% of questions from B have incorrect solutions. Let the following events be defined as

W: solution to a problem is incorrect

S: teacher C thinks solution is incorrect

E_1: Problem is formed by A

E_2 : Problem is formed by B

On the basis of above information answer the following:

58. The Probability that solution to a randomly selected problem is correct, i.e. $P(\overline{W})=$

A) $\frac{5}{6}$ **B)** $\frac{1}{6}$ **C)** $\frac{17}{20}$ **D)** $\frac{2}{3}$

59. The probability that teacher C thinks a given problem has correct solution, i.e. $P(\overline{S})=$

A) $\frac{4}{15}$ **B)** $\frac{11}{5}$ **C)** $\frac{13}{15}$ **D)** $\frac{2}{15}$

60. If solution of a problem randomly selected is definitely correct, then the probability that it was formed by A, i.e. $P\left(\frac{E_1}{\overline{W}}\right)$ is equal to

A) $\frac{9}{10}$ **B)** $\frac{4}{5}$ **C)** $\frac{1}{3}$ **D)** $\frac{9}{25}$

Passage – 2

Conside an equation $z^{12}+z^6+1=0$, whose roots are $\alpha_1,\alpha_2.....\alpha_{12}$ and $\theta_1,\theta_2.....\theta_{12}$ are the arguments of the roots of the equation, where $0<\theta_1<\theta_2<.....<\theta_{12}<2\pi$.

On the basis of above information answer the following:

61. $\cos\left(\sum_{i=1}^{12}\theta_i\right)+i\sin\left(\sum_{i=1}^{12}\theta_i\right)$ is equal to

A) 0 **B)** -1 **C)** 1 **D)** $\frac{1}{2}$

62. Sum of real parts of all roots of the equation whose imaginary part is negative, is

A) $\cos 10^{\mathbf{o}}$ **B)** $\cot 10^{\mathbf{o}}$ **C)** $-\frac{\cos^2 10^{\mathbf{o}}}{\sin 10^{\mathbf{o}}}$ **D)** 0

SECTION –IV

(INTEGER ANSWER TYPE)

This section contains **7 questions.** The answer to each of the questions is a **single-digit integer,** ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS

63. PABC is a tetrahedron such that $\angle APB = \angle BPC = \angle CPA = 90^0$. If the sum of its six edges is constant S and let maximum volume of the tetrahedron be $\frac{5\sqrt{2}-7}{k}S^3$, then sum of digit(s) of k is equal to

64. Consider the sets $(m, c \in R)$:

$A = (x, y) : x^2 + y^2 - 2|x| - 2|y| + 1 = 0$

and $B = (x, y) : y = mx + c$

i)If $A \cap B$ is a non-empty set for all real values of m, then the number of integral values(s) of c are p

ii) If $A \cap B$ has five elements for $m = 1$, then the number of values of c are q .Then the value of $p + q$ is

65. Let a_n denotes the number of sequences of length n formed by using the digits 0,1 such that there are no two consecutive 1 in the sequence, then the value of

$$\left(\left(\frac{a_5 a_6 - {a_5}^2}{a_4}\right) - 5\left(\frac{a_n - a_{n-1}}{a_{n-2}}\right)\right)$$ is equal to

(where 010,001,100,101, all are sequences of length three)

66. Let the line $y = mx$ and $\log(x^2 + y^2) = 2\cot\sqrt{\frac{\pi^2}{16} - \alpha^2}\tan^{-1}\left(\frac{y}{x}\right)$ intersect each other at an acute angle θ. If maximum value of θ is equal to $\frac{b\pi}{4}$, then b is equal to

67. Let $ax^4 + bx^3 + x^2 + (3-a)x + 3 = 0$ and $x^2 + (2-a)x + 3 = 0$ have common roots. If $a \in (-1,5)$, then find $2|a - 12b|$

68. A real tangent to the hyperbola $\frac{x^2}{4} - y^2 = 1$ meets the ellipse $x^2 + 4y^2 = 4$ in two distinct point P & Q. Then total number of points where locus of mid-point of chord PQ meets x-axis is/are equal to

69. The number of functions f, from the set $A = \{x \in N : x^2 - 10x + 9 \le 0\}$ to the set $B = \{n^2 : n \in N\}$ such that $f(x) \le (x-3)^2 + 1$, for every $x \in A$, is K, then the sum of digit(s) of K is